

7. Alternatives to the Global Search

Outline

- ① Classification in IR (manual process)
- ② Cluster Approaches (Automatic Classification)
- ③ Browsing (Exploratory Search)

7.1 Classification in IR

- Classification: structuring of a difficult to understand set of objects (SQL: group by)
- Definition according to Gabler's business informatics lexicon:
 - ▶ “**Classification**, abstraction concept leading to a division of things and concepts linked by common characteristics. From both a scientific and a practical point of view, classification serves to increase the manageability of any field of work.”

Example of classification

Set of students classified by “mother tongue” and stage of study

- French in basic studies
- German in basic studies
- Italian in main studies
- Austrian in main studies

Example shows strong relation to a concrete application and that classes are not necessarily disjoint

Goal of classification

- Structuring the elements of a population according to **features**
- Usually alternative features possible
- Example:
 - ▶ division of documents according to *“monographs”, “anthologies”, “journal articles”, ...*
 - ▶ alternative features: *authors, number of pages, year of publication, topics covered, ...*

Classification and Information Retrieval

- Classification by topics
- However, very high diversity of topics
⇒ multilevel approach, topic hierarchies or trees
- Use of classification scheme in libraries
goal: find quickly related titles
- Other popular examples:
 - ▶ groups of substances in chemistry, taxonomies in biology
 - ▶ subject classification in encyclopedias

TARIC (customs tariffs of the EU)

SECTION XV base metals and articles of base metal	
SECTION XVI machinery and mechanical appliances; electrical equipment; parts thereof; sound recorders and reproducers, television image and sound recorders and reproducers, and parts and accessories of such articles	
Chapter 84	nuclear reactors, boilers, machinery and mechanical appliances; parts thereof
Chapter 85	electrical machinery and equipment and parts thereof; sound recorders and reproducers, television image and sound recorders and reproducers, and parts and accessories of such articles
SECTION XVII vehicles, aircraft, vessels and associated transport equipment	8522 Parts and accessories suitable for use solely or principally with sound reproducing and recording apparatus and with video equipment for recording and reproducing pictures and sound
SECTION XVIII optical, photographic, cinematographic, measuring, checking, precision, medical or surgical instruments and apparatus; clocks and watches; musical instruments; parts and accessories thereof	85221000 Pick-up cartridges
	85229000 Parts and accessories suitable for use solely or principally with sound reproducing and recording apparatus and with video equipment for recording and reproducing pictures and sound (excl. pick-up devices for grooved recording media)

International Patent Classification (IPC)

- Nine-volume manual for the classification of filed and accepted patents
- Hierarchical system
- Levels:
 - ▶ sections
 - ▶ classes
 - ▶ subclasses
 - ▶ groups
 - ▶ subgroups

Problems of classifications

- Innovative topics often not considered
- High effort for setup and maintenance
- No clear classification of interdisciplinary topics

Sections

Section	Content
A	daily living needs
B	working method; transporting
C	chemistry; metallurgy
D	textiles; paper
E	construction; earth drilling; mining
F	mechanical engineering; lighting; heating; weapons; blasting
G	physics
H	electrical engineering

Example path

Key	Level	Designation
G	section	physics
G01	class	measure; test
G01R	subclass	measurement of electrical quantities; measurement of magnetic quantities
G01R 23	group	arrangements for measuring frequencies; arrangements for analyzing frequency spectra
G01R 23/16	subgroup	Spectral analysis; Fourier analysis

Revisions of the IPC

- Problem: innovations often do not fit well into classification
- Therefore, every 5 years revision
- Previously classified patents remain in old version
- Retrieval must therefore take into account older revisions

ACM Classification System

- **A:** General Literature
 - **A.0:** GENERAL
 - **A.1:** INTRODUCTORY AND SURVEY
 - **A.2:** REFERENCE (e.g., dictionaries, encyclopedias, glossaries)
 - **A.m:** MISCELLANEOUS
- **B:** Hardware
 - **B.0:** GENERAL
 - **B.1:** CONTROL STRUCTURES AND MICROPROGRAMMING
 - **B.1.0:** General
 - **B.1.1:** Control Design Styles
 - **B.1.2:** Control Structure Performance Analysis and Design Aids
 - **B.1.3:** Control Structure Reliability, Testing, and Fault-Tolerance
 - **B.1.4:** Microprogram Design Aids
 - **B.1.5:** Microcode Applications
 - **B.1.m:** Miscellaneous
 - **B.2:** ARITHMETIC AND LOGIC STRUCTURES
 - **B.2.0:** General
 - **B.2.1:** Design Styles
 - **B.2.2:** Performance Analysis and Design Aids
 - **B.2.3:** Reliability, Testing, and Fault-Tolerance
 - **B.2.4:** High-Speed Arithmetic
 - **B.2.m:** Miscellaneous
 - **B.3:** MEMORY STRUCTURES
 - **B.3.0:** General
 - **B.3.1:** Semiconductor Memories
 - **B.3.2:** Design Styles
 - **B.3.3:** Performance Analysis and Design Aids
 - **B.3.4:** Reliability, Testing, and Fault-Tolerance
 - **B.3.m:** Miscellaneous
 - **B.4:** INPUT/OUTPUT AND DATA COMMUNICATIONS
 - **B.4.0:** General
 - **B.4.1:** Data Communications Devices
 - **B.4.2:** Input/Output Devices
 - **B.4.3:** Interconnections (Subsystems)
 - **B.4.4:** Performance
 - **B.4.5:** Reliability, Testing, and Fault-Tolerance
 - **B.4.m:** Miscellaneous
 - **B.5:** REGISTER-TRANSFER LOGIC
 - **B.5.0:** General
 - **B.5.1:** Design
 - **B.5.2:** Design Aids
 - **B.5.3:** Reliability, Testing, and Fault-Tolerance
 - **B.5.m:** Miscellaneous
 - **B.6:** LOGIC DESIGN

Categories and Subject Descriptors

H.3.3 [Information Storage and Retrieval]: Information Search and Retrieval—*Relevance feedback* Retrieval models
 Selection process Information filtering; H.3.4 [Information Storage and Retrieval]: Systems and Software—*Performance evaluation* (efficiency and effectiveness)

The decimal classification

- Universal Decimal Classification (UDC)
- Unification in the library system
- Developed in 1876 by the American librarian Malvil Dewey (1851-1931)
- Each basic science identified by a digit, such as 0 for general, 1 for philosophy and psychology
- Subgroups identified by next digit and so on

Example classification group 622.33

- 6 for Applied Sciences, Medicine, Technology
- 2 for Engineering, Technology in general
- 2 for Mining
- 3 for Mining of specific minerals, ores, rocks
- 3 for Coal mining

Groups of 3 are separated from each other by a dot

Faceting

- UDC auxiliary tables

- ▶ division of the documents according to place, time, form, language or finer subject classification
- ▶ noted in parentheses

Examples

- Place: 622(430) for “Mining Technology in Germany”
- Symmetrical relationship between 2 topics: 621.3:622 the “Electrical Engineering in Mining”
- Relationship: 622+629 classified “Mining and Metallurgy”
- Interval with “/”: [23/28:294.3](540) classifies “Christianity in relation of Buddhism in India”

General attachment numbers

The following preceding or enclosing characters are permitted:

- = language
- (0...) form
- (...) place
- (=...) human ancestry, ethnic grouping and nationality
- "... " time
- .00 viewpoint
- -05 persons and personal characteristics

Linking decimal classifications

- + enumeration of several facts
- : symmetrical relationship between two facts
- :: asymmetrical relationship between two facts
- / extension character for grouping together several adjacent decimal classification numbers
- ' grouping character to form new facts from the combination of individual decimal classification components

Catalogs on the Internet

- 2 approaches to search the Internet:
 - ▶ search engines like *Google*
 - ▶ catalogs like *Yahoo*, *Web.de*
- Both approaches work mostly in parallel
- Similar to librarianship
- But spatial ordering of physical documents is not required
⇒ multiple entries possible
- Manual maintenance of the classification required

Types of classification hierarchies

- Monoclassification hierarchy: a class is subordinate to only one other class
⇒ tree
- Polyclassification hierarchy: subordination to multiple classes
⇒ hierarchy

Problems with classifications

- Fast aging due to manual creation
- Therefore inhomogeneities are unavoidable
- Goal: monoclassification if possible

7.2 Clustering Approaches

- Automated division of a set into subsets (**clusters**)
 - ▶ elements within clusters are similar
 - ▶ elements of different clusters are dissimilar
- Large sets get structured
- Undiscovered relationships become visible

Clustering analysis and Information Retrieval

Example applications:

- ① Clustering documents based on their representations:
 - ▶ on entire collection or
 - ▶ on result set
- ② Clustering of documents based on other criteria, such as citations
- ③ Clustering of terms to find synonyms

Similarity measures and distances

- 1 Similarity or distance (such as Euclidean distance) between documents
- 2 Document length should not have any influence

Cosine measure

Sum over product of all terms
appears again

$$sim_{D_i, D_j} = \frac{\sum_{k=1}^t (w_{ik} \cdot w_{jk})}{\sqrt{\sum_{k=1}^t w_{ik}^2 \cdot \sum_{k=1}^t w_{jk}^2}}$$

Normalization

Dice coefficient

$$sim_{D_i, D_j} = \frac{2 \cdot \sum_{k=1}^t (w_{ik} \cdot w_{jk})}{\sum_{k=1}^t w_{ik}^2 + \sum_{k=1}^t w_{jk}^2}$$

Jaccard coefficient

$$sim_{D_i, D_j} = \frac{\sum_{k=1}^t (w_{ik} \cdot w_{jk})}{\sum_{k=1}^t w_{ik}^2 + \sum_{k=1}^t w_{jk}^2 - \sum_{k=1}^t (w_{ik} \cdot w_{jk})}$$

$$J(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}$$

Similarity matrix

- Similarity measures generate a symmetric similarity matrix
- Save as inverted list

$$SIM = \begin{pmatrix} sim_{D_2,D_1} & & & & \\ sim_{D_3,D_1} & sim_{D_3,D_2} & & & \\ sim_{D_4,D_1} & sim_{D_4,D_2} & sim_{D_4,D_3} & & \\ \vdots & \vdots & \vdots & \ddots & \\ sim_{D_N,D_1} & sim_{D_N,D_2} & sim_{D_N,D_3} & \cdots & sim_{D_N,D_{N-1}} \end{pmatrix}$$

Clustering algorithms

- Hierarchical algorithms
- Non-hierarchical algorithms

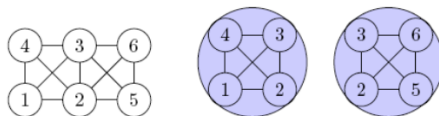
Non-hierarchical clustering approaches

Variants

- Cluster **generation**
- Cluster **update**

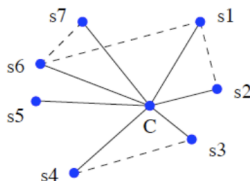
Clique technique as generating non-hierarchical algorithm

- Each element of a cluster has similarity over a given threshold to every other element of the cluster
- Maximum clusters (cliques) are searched for
- Problems:
 - ▶ an element can belong to several clusters
 - ▶ cluster can contain only one element
 - ▶ expensive calculation



Star technique as generating non-hierarchical algorithm

- Differentiation of satellite and cluster nodes
- Cluster node forms cluster with satellite nodes being more similar to cluster node than a threshold
- One element can appear in several clusters
- Quadratic costs



Reallocation technique as updating, non-hierarchical algorithm

- Assignment of new element to the best cluster
 - ▶ centroid (artificial average element) is managed for each cluster
 - ▶ calculation of similarity to each centroid
 - ▶ assignment to the most similar centroid
 - ▶ recalculation of the centroids
 - ▶ reassignment of all elements regarding new centroid
 - ▶ centroids must then be adjusted again ...
- Repeat until stability achieved

Determination of the threshold

- Too high $\Rightarrow$ single element cluster
- Too low $\Rightarrow$ very extensive clusters
- Hierarchical algorithms are better in that respect

Hierarchical clustering approaches

- **Variants**

- ▶ **agglomerative** algorithms: successive merging of single element clusters
 - ▶ **dividing** algorithms: successive decomposition of N-element clusters into smaller clusters
- Advantage: relatively low computational effort, since number of cluster variants is reduced step by step
- Disadvantage: once a decision have been made, it cannot be revised

Agglomerative algorithms

Variants of how clusters are merged

- Single link
- Complete link
- Average link
- Centroid method
- Ward's method

Single link

- Clusters with the greatest similarity are grouped together
- Similarity between clusters as **maximum** similarity between their elements

$$\text{MAX}_{x \in A; y \in B} \text{sim}(x, y)$$

- Disadvantage: large clusters tend to increase in size

Complete Link

- Clusters with the greatest similarity are grouped together
- Similarity between clusters as **minimal** similarity between their elements

$$MIN_{x \in A; y \in B} sim(x, y)$$

- small clusters are preferred for merging

Average link

- Clusters with the greatest similarity are grouped together
- Similarity between clusters as **average** similarity between their elements

$$\frac{1}{|A \times B|} \cdot \sum_{x \in A; y \in B} \text{sim}(x, y)$$

- Compromise between single and complete link

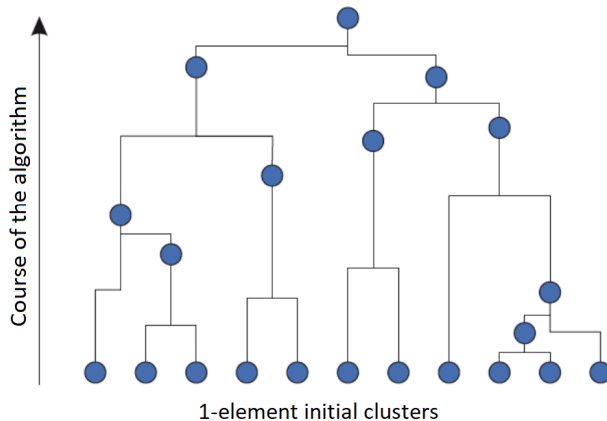
Centroid link

- Clusters with the greatest similarity are grouped together
- Similarity between clusters as similarity between their centroids

Ward's method

- Clusters with smallest **error sum** are merged
- Error sum is squares of individual Euclidean distances of cluster elements
- Advantage: homogeneous clusters and symmetrical cluster hierarchy
- Disadvantage: high calculation effort

Agglomerative hierarchical clustering algorithms: dendrogram



General agglomerative algorithm

Basis: similarity matrix

- ① Start with N one-element clusters (numbered accordingly)
- ② Search for pair (x, y) with highest similarity
- ③ Merge x and y :
 - ① merge the columns and rows of x and y
 - ② new value is maximum for single link (analog for the other algorithms)
 - ③ new cluster gets smaller number from the original clusters
- ④ Steps are repeated $N - 1$ times

Example average link

$$\frac{1}{|A \times B|} \cdot \sum_{x \in A; y \in B} \text{sim}(x, y)$$

How are the new similarity values calculated?

		Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6
		$\{D_1\}$	$\{D_2\}$	$\{D_3\}$	$\{D_4\}$	$\{D_5\}$	$\{D_6\}$
Cluster 1	$\{D_1\}$	∞	4	0	13	8	16
Cluster 2	$\{D_2\}$	4	∞	3	2	9	3
Cluster 3	$\{D_3\}$	0	3	∞	0	4	5
Cluster 4	$\{D_4\}$	13	2	0	∞	8	20
Cluster 5	$\{D_5\}$	8	9	4	8	∞	3
Cluster 6	$\{D_6\}$	16	3	5	20	3	∞



Merging of clusters 4 and 6

		Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
		$\{D_1\}$	$\{D_2\}$	$\{D_3\}$	$\{D_4, D_6\}$	$\{D_5\}$
Cluster 1	$\{D_1\}$	∞	4	0	14,5	8
Cluster 2	$\{D_2\}$	4	∞	3	2,5	9
Cluster 3	$\{D_3\}$	0	3	∞	2,5	4
Cluster 4	$\{D_4, D_6\}$	14,5	2,5	2,5	∞	5,5
Cluster 5	$\{D_5\}$	8	9	4	5,5	∞

Merging of clusters 1 and 4

$$\frac{1}{|A \times B|} \cdot \sum_{x \in A; y \in B} \text{sim}(x, y)$$

		Cluster 1	Cluster 2	Cluster 3	Cluster 5
		$\{D_1, D_4, D_6\}$	$\{D_2\}$	$\{D_3\}$	$\{D_5\}$
Cluster 1	$\{D_1, D_4, D_6\}$	∞	3	1,67	6,33
Cluster 2	$\{D_2\}$	3	∞	3	9
Cluster 3	$\{D_3\}$	1,67	3	∞	4
Cluster 5	$\{D_5\}$	6,33	9	4	∞

Merging of clusters 2 and 5

		Cluster 1	Cluster 2	Cluster 3
		$\{D_1, D_4, D_6\}$	$\{D_2, D_5\}$	$\{D_3\}$
Cluster 1	$\{D_1, D_4, D_6\}$	∞	4,67	1,67
Cluster 2	$\{D_2, D_5\}$	4,67	∞	3,5
Cluster 3	$\{D_3\}$	1,67	3,5	∞

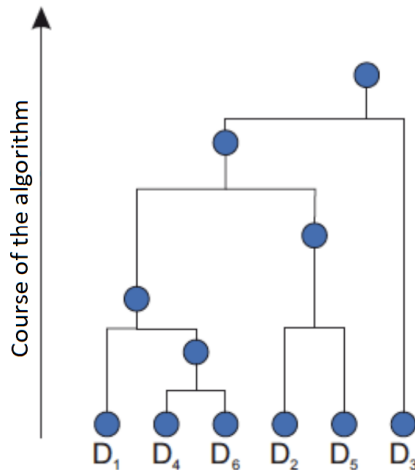
Merging of clusters 1 and 2

		Cluster 1	Cluster 3
		$\{D_1, D_4, D_6, D_2, D_5\}$	$\{D_3\}$
Cluster 1	$\{D_1, D_4, D_6, D_2, D_5\}$	∞	2,4
Cluster 3	$\{D_3\}$	2,4	∞

Merging the remaining clusters

		Cluster 1
		$\{D_1, D_4, D_6, D_2, D_5, D_3\}$
Cluster 1	$\{D_1, D_4, D_6, D_2, D_5, D_3\}$	∞

Dendrogram



Application areas of clustering algorithms

- Automatic thesaurus generation
- Search along a cluster hierarchy
- Preparation of the result of a query
- Maintenance of a classification

Automatic thesaurus generation

- Thesaurus in order to increase recall value
- Instead of manually created thesaurus: **term clustering**
- Application of star technique
- Starting point: **term $\times$ document matrix**, occurrence frequencies in the cells

Example: term $\times$ document matrix

	D_1	D_2	D_3	D_4	D_5	D_6	D_7
t_1	0	4	0	3	0	1	0
t_2	3	0	4	0	0	0	0
t_3	0	2	0	2	0	0	0
t_4	1	0	0	4	0	0	0
t_5	2	0	2	0	3	0	0
t_6	0	3	0	0	3	0	2
t_7	2	0	7	0	0	0	0
t_8	0	4	0	4	0	0	0
t_9	0	0	0	0	3	0	6
t_{10}	0	1	0	0	2	0	4

Example: term $\times$ term similarity

Here numerator of the dice coefficient and threshold 40

	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
t_1	52	0	28	24	0	24	0	56	0	8
t_2	0	50	0	6	28	0	68		0	0
t_3	28	0	16	16	0	12	0		32	0
t_4	24	6	16	34	4	0	4	32	0	0
t_5	0	28	0	4	34	18	36	0	18	12
t_6	24	0	12	0	18	44	0	24	42	34
t_7	0	68	0	4	36	0	106	0	0	0
t_8	56		32	32	0	24	0	64	0	8
t_9	0		0	0	18	42	0	0	90	60
t_{10}	8		0	4	0	12	34	0	8	60

Example: term $\times$ document matrix

	D_1	D_2	D_3	D_4	D_5	D_6	D_7
t_1	0	4	0	3	0	1	0
t_2	3	0	4	0	0	0	0
t_3	0	2	0	2	0	0	0
t_4	1	0	0	4	0	0	0
t_5	2	0	2	0	3	0	0
t_6	0	3	0	0	3	0	2
t_7	2	0	7	0	0	0	0
t_8	0	4	0	4	0	0	0
t_9	0	0	0	0	3	0	6
t_{10}	0	1	0	0	2	0	4



Search along a cluster hierarchy

- Navigation from root to leaves along a dendrogram
- Each cluster is represented by a typical document (most similar to centroid)
- Advantage: query does not have to be formulated (**browsing**)

Preparation of the result of a query

- Structuring of a query result without sorting (Boolean retrieval) by forming thematic clusters
- Meaningful for result with ranking to separate content orientations
- Problem: no explicit description of the clusters
- Better: classification, but high manual effort

Maintenance of a classification

- Assignment of a new document to a classification using clustering algorithm, such as single link
- Second step: reallocation technique for checking the current assignment
- Test whether classification is conclusive (using similarity matrix)

7.3 Browsing

- **Browsing:** navigational access to information
 - ▶ other aspects: combing, skimming
- Requires appropriate preparation of the document collection
 - ▶ bibliographies and references
 - ▶ classifications
 - ▶ hypertext documents
 - ▶ clusters and cluster hierarchies
 - ▶ visual representation of document collections

Bibliographies and references

- Scientific papers contain bibliographies and references
- Effort to track a reference is high

Classification

- Considering documents in the same class
- Class hierarchy: consider more specific or more general documents
- Browsing from root to leaves

Hypertext documents

- Hypertext formats like HTML contain references to other locations in the document or to other documents
- No clearly defined structure, but a network of references
- High risk of getting lost, therefore mostly in combination with search engine search and catalogs

Clusters and cluster hierarchies

- Considering documents in the same cluster
- Search along cluster hierarchy

Visual representation of document collections

- Idea: complex relationships are easier to comprehend graphically than textually
- Use of navigation tools in the document space
- Information visualization problem

1-dimensional linear representation

- Sorted list of documents
- Sorting for example by classification or publication date

Example: Java API documentation

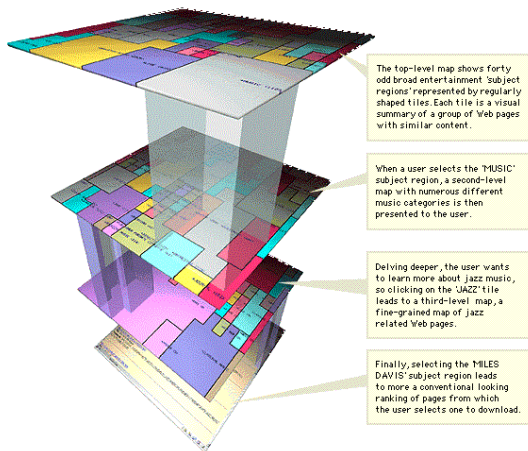
The screenshot shows the Java Platform, Standard Edition 7 API Specification page. The left sidebar contains a navigation pane with 'All Classes' and 'Packages' sections. The 'Packages' section lists various Java packages, including java.applet, java.awt, java.awt.color, java.awt.dnd, java.awt.event, java.awt.font, java.awt.geom, java.awt.im, java.awt.image, and java.awt.image.renderable. The main content area displays the 'Packages' section, listing various Java packages with their descriptions. The packages listed are:

Package	Description
java.applet	Provides the classes necessary to create an applet and the classes an applet uses to communicate with its applet context.
java.awt	Contains all of the classes for creating user interfaces and for painting graphics and images.
java.awt.color	Provides classes for color spaces.
java.awt.datatransfer	Provides interfaces and classes for transferring data between and within applications.
java.awt.dnd	Drag and Drop is a direct manipulation gesture found in many Graphical User Interface systems that provides a mechanism to transfer information between two entities logically associated with presentation elements in the GUI.
java.awt.event	Provides interfaces and classes for dealing with different types of events fired by AWT components.
java.awt.font	Provides classes and interface relating to fonts.
java.awt.geom	Provides the Java 2D classes for defining and performing operations on objects related to two-dimensional geometry.
java.awt.im	Provides classes and interfaces for the input method framework.
java.awt.im.spi	Provides interfaces that enable the development of input methods that can be used with any Java runtime environment.
java.awt.image	Provides classes for creating and modifying images.
java.awt.image.renderable	Provides classes and interfaces for producing rendering-independent images.
java.awt.print	Provides classes and interfaces for a general printing API.
java.beans	Contains classes related to developing beans - components based on the JavaBeans™ architecture.
java.beans.beancontext	Provides classes and interfaces relating to bean context.
java.io	Provides for system input and output through data streams, serialization and the file system.
java.lang	Provides classes that are fundamental to the design of the Java programming language.
java.lang.annotation	Provides library support for the Java programming language annotation facility.
java.lang.instrument	Provides services that allow Java programming language agents to instrument programs running on the JVM.
java.lang.invoke	The java.lang.invoke package contains dynamic language support provided directly by the Java core class libraries and virtual machine.
java.lang.management	Provides the management interfaces for monitoring and management of the Java virtual machine and other components in the Java runtime.
java.lang.ref	Provides reference-object classes, which support a limited degree of interaction with the garbage collector.
java.lang.reflect	Provides classes and interfaces for obtaining reflective information about classes and objects.
java.math	Provides classes for performing arbitrary-precision integer arithmetic (BigInteger) and arbitrary-precision decimal arithmetic (BigDecimal).
java.net	Provides the classes for implementing networking applications.
java.nio	Defines buffers, which are containers for data, and provides an overview of the other NIO packages.
java.nio.channels	Defines channels, which represent connections to entities that are capable of performing I/O operations, such as files and sockets, defines selectors, for multiplexed, non-blocking I/O operations.
java.nio.channels.spi	Service-provider classes for the java.nio.channels package.

2-dimensional representation

- Arrangement in a plane
- Arrangement variants
 - ▶ dimension reduction to 2 dimensions
 - ▶ representation of cluster hierarchies (in combination with dimension reduction)
 - ▶ arrangement based on forces of attraction and repulsion

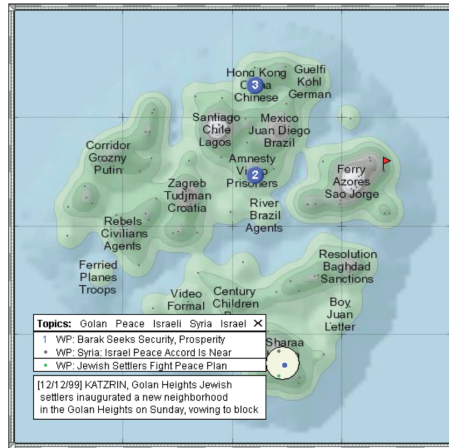
Example: ET-Map



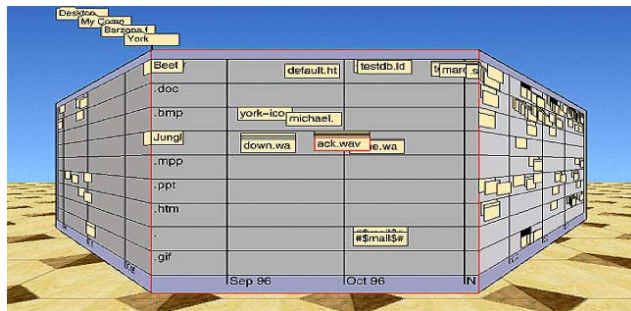
3-dimensional representation

- Further dimension, such as contour lines
- Height can mean number of documents of the topic (NewsMap)
- Time dimension (perspective wall)
- Representation of additional dimensions coded by icons, size and shape of objects

NewsMap



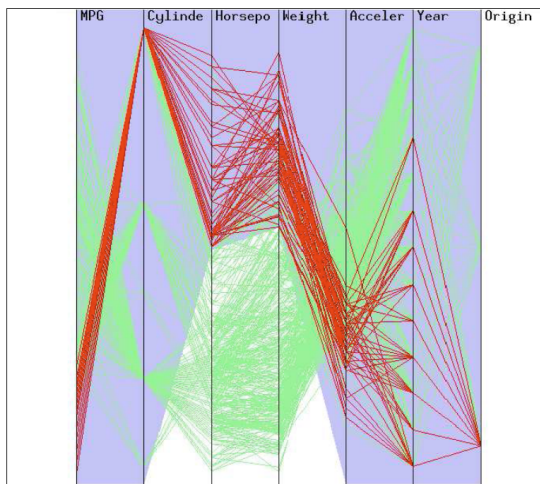
Perspective wall



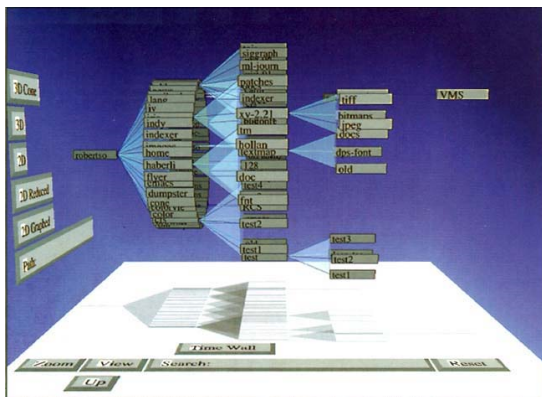
Multidimensional representation

- Additional dimensions represented by icons, colors, size of symbols
- More than 7 dimensions are not useful
- Approach by Matt Ward (criteria of motor vehicles)

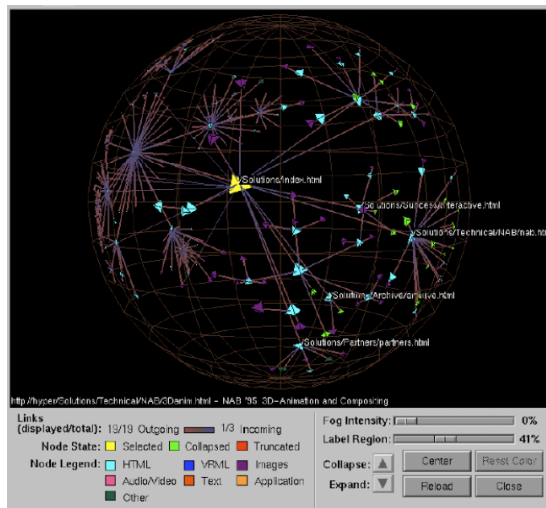
Criteria of motor vehicles



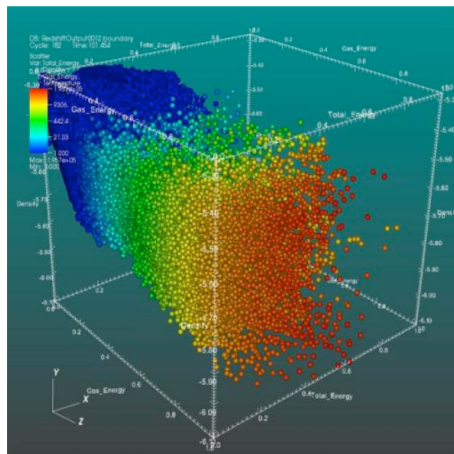
Tree like representation



Representation as a network



Scatter plot



Basics of document collection visualization

- “visual-information-seeking-mantra” by Ben Shneiderman: “Overview first, zoom and filter, then detail on demand”
- Embedding of documents or groups in 1-, 2-, or multidimensional vector space based on similarity or distance as far as possible

Information visualization tasks

Task	Description
Overview	Give an overview of the entire collection.
Zoom	Display the elements of current interest enlarged (in more detail).
Filter	Hide elements that are not currently of interest.
Details-on-Demand	Allow to select an element or a group and show the related detailed information.
Relate	Show the connections and links between the elements (e.g. content references or citations).
History	The previous actions should be saved in order to perform operations such as <i>Undo</i> , <i>Replay</i> or <i>proactive refinement</i> .
Extract	Allow extraction of partial collections or query parameters.